CATARACT

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Cataract = #1 cause of blindness in India!
Super high yield topic!

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Definition, Classification + Etiology

Cataract = opacity of the crystalline lens. Simple but powerful!

DEFINITION

"Any opacity in the crystalline lens or its capsule, irrespective of effect on vision." — Khurana

Classification by Age of Onset

Type	Details
Congenital	Present at birth — rubella, galactosemia, Down syndrome
Developmental	Develops in childhood — lamellar cataract most common
Acquired / Senile	Most common overall (>90%). Age >50 years
Traumatic	Blunt/penetrating — rosette cataract, Vossius ring
Complicated	Secondary to uveitis, glaucoma, retinal detachment
Metabolic	Diabetes (snowflake), galactosemia (oil droplet), hypocalcemia
Drug-induced	Steroids (posterior subcapsular), miotics, amiodarone
Radiation	IR (glassblower's), UV, ionizing radiation

Morphological Classification (Senile)

Type	Location	Key Feature
Nuclear	Central nucleus	Gradual, myopic shift, brown/black
Cortical	Cortex (peripheral)	Spoke-like opacities, glare
Posterior Subcapsular	Posterior capsule	Rapid, near vision loss first

Congenital Cataract — Quick Hits

- Lamellar/Zonular = most common TYPE | Idiopathic = most common CAUSE
- Rubella = most common INFECTIVE cause | Oil droplet = Galactosemia (pathognomonic)
- Vossius ring = iris pigment imprint on lens after blunt trauma (NOT congenital!)

Senile Cataract — Stages + Features

The 4 stages of senile cataract are asked in EVERY single exam!

Stages of Cortical Senile Cataract

Stage	Iris Shadow	Vision	Key Features
Immature	Present	Decreased	Partial opacity, lens swells, shallow AC
Mature	Absent	PL only	Complete opacity, pearly white, normal AC
Hypermature	Absent	May improve!	Shrunken lens, wrinkled capsule, deep AC
Morgagnian	Absent	PL/PR only	Liquefied cortex, nucleus sinks to bottom

★ **Iris Shadow Test: In immature cataract, oblique light creates a shadow on the** opaque lens BEHIND the clear portion. In mature — no shadow (fully opaque).

Clinical Features (Symptoms)

- **Painless progressive loss of vision** — Most important symptom — gradual onset
- **Glare / photophobia** — Especially posterior subcapsular type
- **Uniocular diplopia / polyopia** — Due to irregular refraction through opacity
- **Myopic shift (Second sight)** — Nuclear cataract — reads without glasses!
- **Colored halos** — Water droplets in lens (diff. from glaucoma)
- **Frequent change of glasses** — Progressive refractive error changes

★ Memory Trick — Second Sight! ★

"Second Sight" = Nuclear sclerosis causes myopic shift = old person can suddenly read without glasses! But it's temporary — cataract still progressing!

Nuclear vs Cortical vs PSC (Comparison)

Feature	Nuclear	Cortical	PSC
Vision	Distance > Near	Peripheral first	Near > Distance
Second sight?	YES (myopic shift)	No	No
Glare	Less	Common	Marked
Associations	Aging	Aging, DM	Steroids, DM, uveitis

Lens-Induced (Phacogenic) Conditions

Condition	Stage	Mechanism
Phacomorphic	Intumescent	Swollen lens pushes iris -> angle closure
Phacolytic	Hypermature	Proteins leak -> macrophages block TM -> OAG

SICS vs Phaco vs ECCE
is THE comparison Q!
Learn this table cold!

Indications for Cataract Surgery

1. Visual: Best corrected VA < 6/18 (WHO) or affecting daily activities
2. Medical: Phacomorphic/phacolytic glaucoma, lens-induced uveitis
3. Optical: To visualize fundus (diabetic retinopathy evaluation)

Surgical Techniques — THE Comparison Table

Feature	SICS	Phacoemulsification	ECCE
Incision	6-7mm scleral tunnel	2.8-3.2mm clear corneal	10-12mm limbal
Sutures	Sutureless	Sutureless	Required
Anesthesia	Local (topical/periblb)	Topical	Local/peribulbar
Astigmatism	Less	Least	Most
Recovery	Fast (1-2 weeks)	Fastest (<1 week)	Slowest (6-8 weeks)
Cost	Low (best for India)	High (machine needed)	Moderate
Best for	Hard nuclei, India	Soft-moderate nuclei	When no phaco/SICS
IOL	Rigid PMMA	Foldable (acrylic)	Rigid PMMA
Complications	Few	Fewest	More (wound related)

★ **SICS is the surgery of choice in India's mass cataract surgery camps!**
Phaco = gold standard worldwide, but SICS = best cost-effective for India.

IOL (Intraocular Lens) Types

IOL Type	Material	Key Feature
Rigid IOL	PMMA	Used in ECCE/SICS, needs larger incision
Foldable IOL	Acrylic/Silicone	Used in phaco, small incision
Multifocal IOL	Acrylic	Reduces need for reading glasses
Toric IOL	Acrylic	Corrects pre-existing astigmatism

★ **IOL Power Calculation**
SRK Formula: $P = A - 2.5L - 0.9K$ (P=power, A=constant, L=axial length, K=keratometry)

Indian Context + NPCBVI

India performs 6M+ cataract surgeries/yr! This comes in exams!

National Programme for Control of Blindness + VI

Started:	1976 (one of India's oldest national health programmes)
Renamed:	NPCBVI in 2017 (added Visual Impairment)
Goal:	Reduce prevalence of blindness to 0.3% by 2025
Target:	Cataract surgical rate of 6000/million/year
Strategy:	Free cataract surgery camps + IOL implantation
Funding:	60:40 (Centre:State); 90:10 for NE states

Blindness in India — Key Statistics

Fact	Data
#1 cause of blindness in India	Cataract (62.6% of all blindness)
#2 cause	Refractive errors (19.7%)
Prevalence of blindness	~0.36% (reduced from 1.1% in 2001-02)
Cataract surgeries per year	~6.5 million (highest globally)
CSR (Cataract Surgical Rate)	~6000 per million per year
WHO definition of blindness	VA < 3/60 in better eye
WHO definition of low vision	VA < 6/18 to 3/60

Post-Operative Complications

EARLY (days-weeks)

- Corneal edema (striate keratopathy)
- Anterior chamber reaction (uveitis)
- Hyphema (blood in AC)
- Endophthalmitis (EMERGENCY!)
- Wound leak / iris prolapse

LATE (weeks-months)

- Posterior capsule opacification (PCO)
- Cystoid macular edema (CME)
- Retinal detachment
- IOL decentration / dislocation
- Secondary glaucoma

★ **After-Cataract (PCO)**

PCO = most common LATE complication. Treated by Nd:YAG laser capsulotomy.

Also called "after-cataract" — it's NOT a new cataract! Elschnig pearls on capsule.

Cataract = guaranteed
20+ marks in ophthal!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Classify cataract. Describe stages of senile cataract with clinical features
- 2. Compare SICS vs Phacoemulsification vs ECCE (table format!)
- 3. Describe management of senile cataract including surgical options & IOL
- 4. Enumerate complications of cataract surgery. Describe endophthalmitis
- 5. Write about NPCBVI — objectives, strategies, achievements

Most Repeated SAQs (5–6 marks)

- 1. Stages of senile cortical cataract (with iris shadow test)
- 2. After-cataract (PCO) — cause & treatment
- 3. Congenital cataract — types, causes, management
- 4. Phacomorphic vs Phacolytic glaucoma
- 5. Second sight phenomenon
- 6. IOL — types and SRK formula

LAQ Answer Blueprint

1	Define	Opacity of crystalline lens (Khurana definition)
2	Classify	Draw classification tree (age + morphology)
3	Stages	4 stages table with iris shadow test
4	Compare	SICS vs Phaco vs ECCE table (ALWAYS!)
5	Complications	Early vs Late complications list
6	Indian	NPCBVI, CSR, blindness statistics

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
#1 cause of preventable blindness in India	Cataract (62.6%)
Most common senile cataract type	Nuclear sclerosis
Most common congenital cataract type	Lamellar/Zonular
Oil droplet cataract	Galactosemia
Sunflower cataract	Wilson disease
Rosette cataract	Concussion (trauma)
Glassblower's cataract	Infrared radiation
Christmas tree cataract	Myotonic dystrophy
Subluxation UP & OUT	Marfan syndrome
Subluxation DOWN & IN	Homocystinuria
Most dreaded post-op complication	Endophthalmitis (S. epidermidis)
Most common late complication	PCO (Nd:YAG laser Rx)

♥ Every cataract surgery gives someone their sight back, Sammy!